Cloud technology

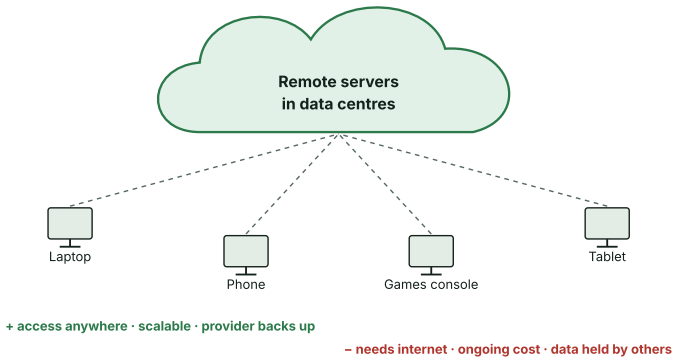
L20 FACT FILE

Specification: students should be able to

- Define the term cloud computing
- Describe the advantages of cloud computing for an organisation
- Describe the disadvantages of cloud computing for an organisation
- Describe the impact of cloud computing on gaming
- Describe the impact of cloud computing on file storage and file sharing, including collaborative tools

CCEA fact file: Fact File 17: Cloud Technology

Key words: cloud computing, remote server, data centre, on demand, subscription, pay as you go, scalability, downtime, bandwidth, carbon footprint, cloud gaming, game streaming, cloud storage, synchronisation, collaboration, version history



Cloud computing: devices reach remote servers over the internet

Cloud computing explained

Cloud computing means using computing resources, such as storage space and software, that are delivered on demand over the internet from remote servers. Instead of keeping files and programs on their own hard drives, organisations and individuals keep them on servers in large data centres owned by cloud providers. They can then reach them from any device with an internet connection. Familiar examples include OneDrive, Google Drive, iCloud and Microsoft 365. Organisations usually pay a monthly subscription or use a pay-as-you-go plan. The cloud is growing quickly because it is flexible: a business can start small and add more storage or users when it needs them. It has also changed everyday life for young people, from saving coursework online to streaming games and working on shared documents with classmates.

Advantages for organisations

For an organisation, the cloud can save money because it does not need to buy, house and maintain its own servers or many separate software licences. Pay-as-you-go plans mean it pays only for what it uses. The service is scalable, so storage and bandwidth can be increased or reduced quickly. Storage capacity is far larger than local hard drives. The provider applies software updates automatically and backs up data at regular intervals, so the organisation does not need its own backup devices or recovery procedures. Staff can access their files from anywhere and on any device, which supports home and hybrid working. If the office suffers a fire, flood or theft, the data is still safe. Using less equipment on site also reduces electricity use, which lowers the organisation's carbon footprint.

Disadvantages for organisations

The biggest weakness of the cloud is that it depends on the internet. If the broadband connection or the router fails, staff cannot reach their files or software. This is called downtime. Downtime can also happen if the provider has a fault or is carrying out maintenance, and the organisation cannot fix the problem itself. Some plans have a bandwidth allowance and charge extra if it is exceeded, and uploading large files can be slow. Security is another concern. The data is stored by a third party, and no provider can promise that it will never be hacked. The organisation is still responsible for protecting personal data under the UK GDPR and the Data Protection Act 2018. Subscription costs continue for as long as the service is used, and getting help from customer support can be slow or cost extra.

Impact on gaming

The cloud has changed gaming for both producers and players. Producers can sell games as downloads, so they save the cost of making discs and paying shops, and can offer lower prices. Players can download a game the moment it is released. Games kept on the producer's servers are much harder to copy illegally than discs, which helps protect sales. The internet also gives producers a worldwide market, and they can release patches, updates and demos straight away. With cloud gaming, also called game streaming, the game runs on powerful servers and video is streamed to the player's device. This means high-quality games can be played on cheap devices without upgrading hardware. However, players need fast broadband with low latency, streaming uses a lot of data, and access is lost if the subscription ends.

Impact on file storage and sharing

Cloud storage keeps files on high-capacity servers. Many apps save work automatically every few seconds, so a crash rarely means lost work. Files remain online until the user deletes them, and the provider keeps copies so it can recover files if a server fails, without the user needing any technical knowledge. Synchronisation means the latest version of a file appears on every device the user owns. Sharing is easier too: instead of emailing large attachments, a user shares a link and sets permissions to decide who can view or edit. Collaborative tools such as Google Docs and Microsoft 365 let several people edit the same document at the same time, wherever they are. Version history shows who made each change and allows an earlier version to be restored, and users can be sent alerts when a document changes.

T Key terms from the mark schemes

Cloud computing
Storing data and running software on remote servers that are accessed on demand over the internet, instead of on a local computer.
`Credited: Using remote servers over the internet`

Data centre
A building containing large numbers of servers that store data and run cloud services.

Pay as you go
A payment model where an organisation pays only for the cloud resources it actually uses.

Cloud gaming
Games run on remote servers and the video is streamed to the player's device over the internet.
`Credited: Game streaming`

Cloud storage
Keeping files on a provider's remote servers so they can be accessed from any device with an internet connection.
`Credited: Online storage`

Scalability
The ability to increase or decrease storage, bandwidth or users quickly as an organisation's needs change.
`Credited: Scalable`

Downtime
A period when a service is unavailable, for example because of an internet failure or provider maintenance.
`Credited: Service outage`

Synchronisation
Keeping copies of a file up to date so that a change made in one place appears on every device and for every user.
`Credited: Sync`

Collaborative tool
Cloud software that lets several people view and edit the same file at the same time, e.g. Google Docs or Microsoft 365.

Carbon footprint
A measure of the impact of an organisation's or person's activities on the environment.

Version history
A record of the changes made to a file that shows who made them and allows earlier versions to be restored.

Exam focus

Cloud questions are usually worth a few marks each but come up often. You may be asked to define cloud computing (2023 Q9, 2024 Q2). A two-mark definition needs two ideas: storing data or running software on remote servers, and accessing them over the internet. Benefits and risks for a business were asked in 2023 Q9 and 2024 Q2. Make each point specific and explain why it matters, for example 'needs an internet connection, so if the broadband fails staff cannot access files'. Cloud gaming was a quality of written communication question in 2019 Q7 and returned in 2023 Q9(c), so practise writing a balanced answer in full sentences that covers both benefits and drawbacks. Cloud storage terms appeared in 2022 Q2. Avoid vague answers such as 'it is cheaper' or 'it is easier' without saying why.

Unit 1 2024 Q2	[4]	Cloud definition, benefits and risks
Unit 1 2023 Q9	[6]	Cloud computing, business benefits, cloud gaming
Unit 1 2022 Q2	[3]	Optical, capacity, cloud storage terms
Unit 1 2019 Q7	[4]	Cloud computing benefits for gaming (QWC)

? Check yourself

1. Define the term cloud computing. [2]
2. A small accountancy firm in Newry keeps all its files on one ageing server in the office. State three benefits the firm would gain by switching to cloud computing. [3]
3. A solicitor's office in Armagh is moving its files to the cloud. Describe two disadvantages of cloud computing for the office. [4]

Answers: 1. Storing data and/or running software/applications (1) on remote servers accessed over the internet (1) / resources delivered on demand (1) – max 2 2. Any three: cost effective / pay as you go / no servers to buy (1) / scalable – storage can be increased or decreased (1) / very large storage capacity (1) / automatic software updates (1) / automatic backup and recovery by the provider (1) / access from anywhere on any device (1) / staff can collaborate on the same files (1) / reduced carbon footprint (1) 3. Two disadvantages, each identified (1) and described in context (1): relies on an internet connection (1) so if the broadband fails staff cannot open client files (1) / client data held by a third party (1) could be hacked, and the office must still meet UK GDPR (1) / provider outages or maintenance (1) mean files are unavailable and the office cannot fix it (1) / ongoing subscription costs (1) that continue every month (1) / bandwidth limits (1) with extra charges if exceeded (1) / slow customer support (1)